

Jenkins Assignment 2

Multiple Release / Patch / Bug-Fix Deployment using Jenkins Freestyle Jobs and Apache

1. Objective

Deploy three independent versions of a website from the same GitHub repository to one EC2 instance using three Jenkins Freestyle jobs. Each release is deployed into a separate Apache directory:

- /var/www/html/R1 → Q1 Release
- /var/www/html/R2 → Q4.2 Patch
- /var/www/html/R3 → Q3.3 Bug Fix

Final browser URLs will be: <http://<EC2-IP>/R1/index.html>, <http://<EC2-IP>/R2/index.html>, and <http://<EC2-IP>/R3/index.html>.

2. Deployment Architecture

Source	Jenkins Job	Apache Target
repo1/index.html	Q1 Release	/var/www/html/R1/
repo2/index.html	Q4.2 Patch	/var/www/html/R2/
repo3/index.html	Q3.3 Bug Fix	/var/www/html/R3/

Screenshot space – deployment architecture / final folder structure:

```
[root@ip-172-31-1-51 R1]# ls -la /var/www/html/R1
total 4
drwxr-xr-x. 2 jenkins jenkins 24 Oct  3 04:56 .
drwxr-xr-x. 5 jenkins jenkins 36 Oct  3 03:19 ..
-rw-r--r--. 1 jenkins jenkins 56 Oct  3 04:56 index.html
[root@ip-172-31-1-51 R1]# ls -la /var/www/html/R2
total 4
drwxr-xr-x. 2 jenkins jenkins 24 Oct  3 04:59 .
drwxr-xr-x. 5 jenkins jenkins 36 Oct  3 03:19 ..
-rw-r--r--. 1 jenkins jenkins 56 Oct  3 05:09 index.html
[root@ip-172-31-1-51 R1]# ls -la /var/www/html/R3
total 4
drwxr-xr-x. 2 jenkins jenkins 24 Oct  3 05:05 .
drwxr-xr-x. 5 jenkins jenkins 36 Oct  3 03:19 ..
-rw-r--r--. 1 jenkins jenkins 56 Oct  3 05:05 index.html
[root@ip-172-31-1-51 R1]#
```

3. Prerequisites

- GitHub repository is available, for example: DV-45/Jenkins-assignment.
- EC2 Linux instance is running.
- Jenkins is installed and accessible.
- Apache HTTP Server is installed and running.
- The EC2 instance can clone the GitHub repository.
- Security Group allows HTTP (TCP 80) and Jenkins access as required.

Check Apache:

```
sudo systemctl status httpd
```

If Apache is not running:

```
sudo systemctl start httpd  
sudo systemctl enable httpd
```

Check the Apache document root:

```
ls -la /var/www/html
```

4. Step 1 – Create Repository Structure

Inside the GitHub repository, create the following structure:

```
Jenkins-assignment/  
└─ jenkins_ec2_assig2/  
    └─ repo1/  
        └─ index.html  
    └─ repo2/  
        └─ index.html  
    └─ repo3/  
        └─ index.html
```

4.1 Create the three HTML files

repo1/index.html

```
<html>  
<head><title>Assignment 2</title></head>  
<body>  
<h1>Hello Q1 Release</h1>  
</body>  
</html>
```

repo2/index.html

```
<html>  
<head><title>Assignment 2</title></head>  
<body>  
<h1>Hello Q4.2 Patch</h1>  
</body>  
</html>
```

repo3/index.html

```
<html>
<head><title>Assignment 2</title></head>
<body>
<h1>Hello Q3.3 Bug Fix</h1>
</body>
</html>
```

Screenshot space – GitHub repository and folder structure:

```
PS C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment> git add -A
PS C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment> git commit -m "Merge remote main into local main"
[main 98ae681] Merge remote main into local main
PS C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment> git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 673 bytes | 673.00 KiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/DV-45/Jenkins-assignment.git
213b519..98ae681 main -> main
PS C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment> cd .\jenkins_ec2_assig2\
PS C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment\jenkins_ec2_assig2> dir

Directory: C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment\jenkins_ec2_assig2

Mode                LastWriteTime         Length Name
----                -
dar--l             01-10-2026   19:41           repo1
dar--l             01-10-2026   19:42           repo2
dar--l             01-10-2026   19:43           repo3

PS C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment\jenkins_ec2_assig2> |
```

4.2 Commit and push

Commit the new folder and push it to GitHub. Example commands:

```
git add jenkins_ec2_assig2/
git commit -m "Add Jenkins Assignment 2 release folders"
git push origin main
```

```

PS C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment> git add -A
PS C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment> git commit -m "Merge remote main into local main"
[main 98ae681] Merge remote main into local main
PS C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment> git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 673 bytes | 673.00 KiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/DV-45/Jenkins-assignment.git
213b519..98ae681 main -> main
PS C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment> cd .\jenkins_ec2_assig2\
PS C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment\jenkins_ec2_assig2> dir

Directory: C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment\jenkins_ec2_assig2

Mode                LastWriteTime         Length Name
----                -
dar--l            01-10-2026      19:41             repo1
dar--l            01-10-2026      19:42             repo2
dar--l            01-10-2026      19:43             repo3
PS C:\Users\darsh\OneDrive\Desktop\GitRepos\Project\Jenkins-assignment\jenkins_ec2_assig2> |

```

5. Step 2 – Prepare EC2 and Apache

On the EC2 instance, confirm Apache is serving from /var/www/html.

```
sudo systemctl status httpd
ls -la /var/www/html
```

Create the deployment directories before running Jenkins jobs:

```
sudo mkdir -p /var/www/html/R1 /var/www/html/R2 /var/www/html/R3
sudo chown -R jenkins:jenkins /var/www/html/R1 /var/www/html/R2 /var/www/html/R3
```

If your Jenkins process uses a different operating-system user, replace jenkins:jenkins with the appropriate user/group. The Jenkins process must have permission to write to R1, R2 and R3.

```

[root@ip-172-31-1-51 html]# ls -la /var/www/html
total 0
drwxr-xr-x. 5 root root 36 Oct  3 03:19 .
drwxr-xr-x. 4 root root 33 Oct  3 03:12 ..
drwxr-xr-x. 2 root root  6 Oct  3 03:19 R1
drwxr-xr-x. 2 root root  6 Oct  3 03:19 R2
drwxr-xr-x. 2 root root  6 Oct  3 03:19 R3

```

6. Step 3 – Create Jenkins Freestyle Job 1 (Q1 Release)

Create a new Jenkins item named: Assignment2-Q1-Release

Select: Freestyle project

6.1 Source Code Management

- Select Git.
- Repository URL: your GitHub repository URL.
- Select the required branch, normally main.
- Add Git credentials if the repository is private.

Important: the repository root contains jenkins_ec2_assig2, so the source path is:

[jenkins_ec2_assig2/repo1](#)

Screenshot space – Job 1 SCM configuration:

The screenshot shows the Jenkins configuration page for a job named "Assignment2-Q1-Release". The left sidebar contains a "Configure" menu with options: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The main content area is divided into several sections:

- General:** Includes a "Description" text area, a "Plain text" preview, and checkboxes for "Discard old builds", "GitHub project", "This project is parameterized", "Throttle builds", and "Execute concurrent builds if necessary". An "Advanced" dropdown is visible.
- Source Code Management:** Features a "Connect and manage your code repository" section with a radio button for "Git" selected. Below it, a "Repositories" section contains a "Repository URL" field with the value "https://github.com/DV-45/jenkins-assignment.git", a "Credentials" dropdown set to "- none -", and an "Add Repository" button. A "Branches to build" section has a "Branch Specifier" field with the value "*/main" and an "Add Branch" button. A "Repository browser" dropdown is set to "(Auto)". An "Additional Behaviours" section has an "Add" button.
- Triggers:** Includes a "Set up automated actions" section with checkboxes for "Trigger builds remotely", "Build after other projects are built", "Build periodically", "GitHub hook trigger for GITScm polling", and "Poll SCM".
- Environment:** Includes a "Configure settings and variables" section with checkboxes for "Delete workspace before build starts", "Use secret text(s) or file(s)", "Add timestamps to the Console Output", "Inspect build log for published build scans", "Terminate a build if it's stuck", and "With Ant".
- Build Steps:** Includes an "Automate your build process" section with a "Execute shell" button. Below it, a "Command" text area contains the following commands:

```
sudo shctl -p /var/www/html/R1
sudo cp -r jenkins_ec2_assig2/repo1/* /var/www/html/R1/
ls -la /var/www/html/R1
```

An "Advanced" dropdown and an "Add build step" button are also present.
- Post-build Actions:** Includes a "Define what happens after a build completes" section with an "Add post-build action" button.

The bottom right corner of the page shows "REST API" and "Jenkins 2.560.1".

6.2 Build Step

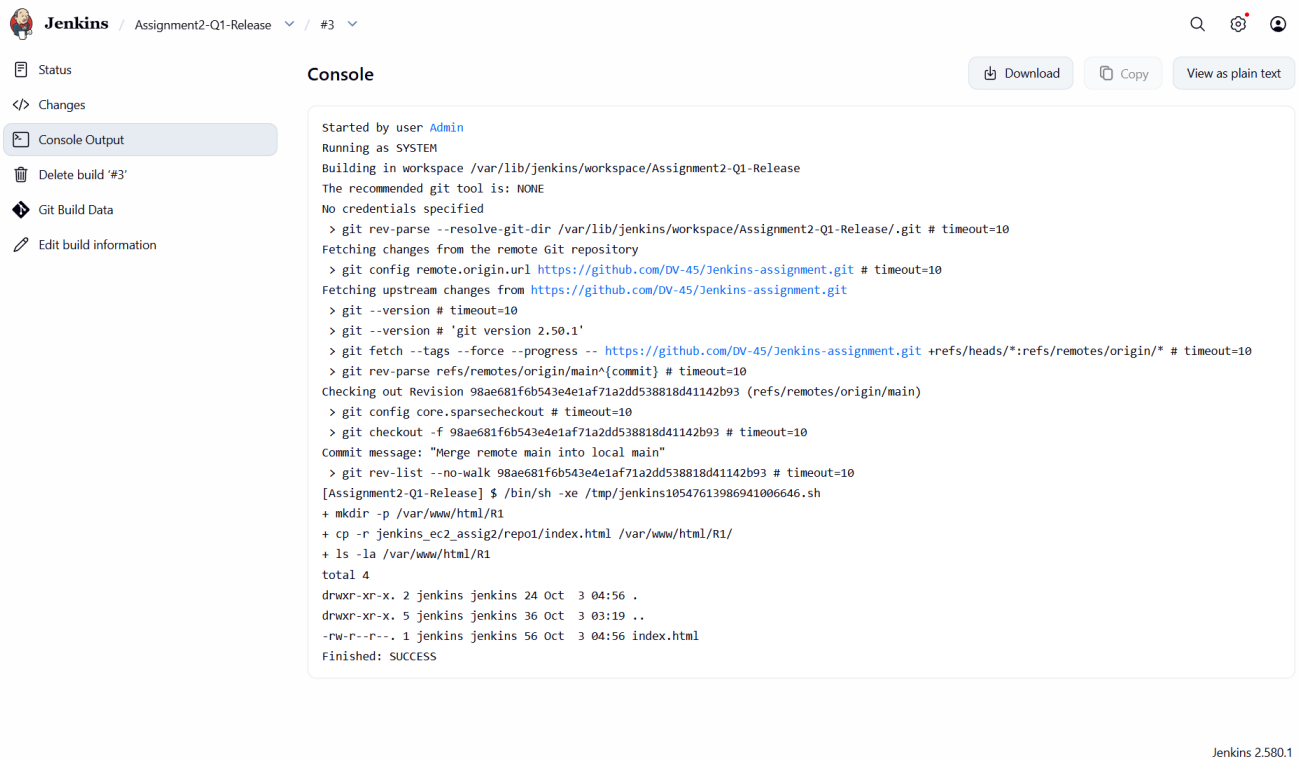
Add Build Step → Execute shell.

Use this command:

```
mkdir -p /var/www/html/R1
cp -r jenkins_ec2_assig2/repo1/* /var/www/html/R1/
ls -la /var/www/html/R1
```

If Jenkins already has write permission on /var/www/html/R1, sudo can be omitted. For a simple lab setup, the important requirement is that the Jenkins build can write to the target directory.

Click Save → Build Now.



The screenshot shows the Jenkins web interface. On the left, there's a sidebar with 'Jenkins' at the top, followed by 'Assignment2-Q1-Release' and '#3'. Below this are links for 'Status', 'Changes', 'Console Output' (which is selected), 'Delete build '#3'', 'Git Build Data', and 'Edit build information'. The main area is titled 'Console' and contains a large text box with the following output:

```
Started by user Admin
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Assignment2-Q1-Release
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/Assignment2-Q1-Release/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/DV-45/Jenkins-assignment.git # timeout=10
Fetching upstream changes from https://github.com/DV-45/Jenkins-assignment.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/DV-45/Jenkins-assignment.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 98ae681f6b543e4e1af71a2dd538818d41142b93 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 98ae681f6b543e4e1af71a2dd538818d41142b93 # timeout=10
Commit message: "Merge remote main into local main"
> git rev-list --no-walk 98ae681f6b543e4e1af71a2dd538818d41142b93 # timeout=10
[Assignment2-Q1-Release] $ /bin/sh -xe /tmp/jenkins10547613986941006646.sh
+ mkdir -p /var/www/html/R1
+ cp -r jenkins_ec2_assig2/repo1/index.html /var/www/html/R1/
+ ls -la /var/www/html/R1
total 4
drwxr-xr-x. 2 jenkins jenkins 24 Oct 3 04:56 .
drwxr-xr-x. 5 jenkins jenkins 36 Oct 3 03:19 ..
-rw-r--r--. 1 jenkins jenkins 56 Oct 3 04:56 index.html
Finished: SUCCESS
```

At the bottom right of the console output area, it says 'Jenkins 2.580.1'.

7. Step 4 – Create Jenkins Freestyle Job 2 (Q4.2 Patch)

Create a new Freestyle project named: Assignment2-Q4.2-Patch

SCM:

- Git repository: same GitHub repository.
- Branch: main.
- Source path used by the build: jenkins_ec2_assig2/repo2.

Build Step → Execute shell:

```
mkdir -p /var/www/html/R2
cp -r jenkins_ec2_assig2/repo2/* /var/www/html/R2/
ls -la /var/www/html/R2
```

Click Save → Build Now.

The screenshot shows the Jenkins web interface. The top bar indicates the job is 'Assignment2-Q4.2-Patch' and it's the first build. The left sidebar has links for Status, Changes, Console Output (selected), Delete build '#1', Git Build Data, and Edit build information. The main console area shows the following output:

```
Started by user Admin
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Assignment2-Q4.2-Patch
The recommended git tool is: NONE
No credentials specified
Cloning the remote git repository
Cloning repository https://github.com/DV-45/Jenkins-assignment.git
> git init /var/lib/jenkins/workspace/Assignment2-Q4.2-Patch # timeout=10
Fetching upstream changes from https://github.com/DV-45/Jenkins-assignment.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/DV-45/Jenkins-assignment.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/DV-45/Jenkins-assignment.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 98ae681f6b543e4e1af71a2dd538818d41142b93 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 98ae681f6b543e4e1af71a2dd538818d41142b93 # timeout=10
Commit message: "Merge remote main into local main"
First time build. Skipping changelog.
[Assignment2-Q4.2-Patch] $ /bin/sh -xe /tmp/jenkins2852939822951588447.sh
+ mkdir -p /var/www/html/R2
+ cp -r jenkins_ec2_assig2/repo2/index.html /var/www/html/R2/
+ ls -la /var/www/html/R2
total 4
drwxr-xr-x. 2 jenkins jenkins 24 Oct 3 04:59 .
drwxr-xr-x. 5 jenkins jenkins 36 Oct 3 03:19 ..
-rw-r--r--. 1 jenkins jenkins 56 Oct 3 05:09 index.html
Finished: SUCCESS
```

At the bottom right of the console area, it says 'Jenkins 2.50.1'.

8. Step 5 – Create Jenkins Freestyle Job 3 (Q3.3 Bug Fix)

Create a new Freestyle project named: Assignment2-Q3.3-Bug-Fix

SCM:

- Git repository: same GitHub repository.
- Branch: main.
- Source path used by the build: jenkins_ec2_assig2/repo3.

Build Step → Execute shell:

```
mkdir -p /var/www/html/R3
cp -r jenkins_ec2_assig2/repo3/* /var/www/html/R3/
ls -la /var/www/html/R3
```

Click Save → Build Now.

Status

Changes

Console Output

Delete build '#2'

Git Build Data

Edit build information

Console

Download

Copy

View as plain text

```

Started by user Admin
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Assignment2-Q3.3-Bug-Fix
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/Assignment2-Q3.3-Bug-Fix/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/DV-45/Jenkins-assignment.git # timeout=10
Fetching upstream changes from https://github.com/DV-45/Jenkins-assignment.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/DV-45/Jenkins-assignment.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 98ae681f6b543e4e1af71a2dd538818d41142b93 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 98ae681f6b543e4e1af71a2dd538818d41142b93 # timeout=10
Commit message: "Merge remote main into local main"
First time build. Skipping changelog.
[Assignment2-Q3.3-Bug-Fix] $ /bin/sh -xe /tmp/jenkins7844763507978401380.sh
+ mkdir -p /var/www/html/R3
+ cp -r jenkins_ec2_assig2/rep03/index.html /var/www/html/R3/
+ ls -la /var/www/html/R3
total 4
drwxr-xr-x. 2 jenkins jenkins 24 Oct 3 05:05 .
drwxr-xr-x. 5 jenkins jenkins 36 Oct 3 03:19 ..
-rw-r--r--. 1 jenkins jenkins 56 Oct 3 05:05 index.html
Finished: SUCCESS

```

Jenkins 2.580.1

9. Step 6 – Verify Apache Deployment

On the EC2 instance, run:

```

ls -la /var/www/html/R1
ls -la /var/www/html/R2
ls -la /var/www/html/R3

```

Expected result:

- /var/www/html/R1/index.html
- /var/www/html/R2/index.html
- /var/www/html/R3/index.html

```

[root@ip-172-31-1-51 R1]# ls -la /var/www/html/R1
total 4
drwxr-xr-x. 2 jenkins jenkins 24 Oct 3 04:56 .
drwxr-xr-x. 5 jenkins jenkins 36 Oct 3 03:19 ..
-rw-r--r--. 1 jenkins jenkins 56 Oct 3 04:56 index.html
[root@ip-172-31-1-51 R1]# ls -la /var/www/html/R2
total 4
drwxr-xr-x. 2 jenkins jenkins 24 Oct 3 04:59 .
drwxr-xr-x. 5 jenkins jenkins 36 Oct 3 03:19 ..
-rw-r--r--. 1 jenkins jenkins 56 Oct 3 05:09 index.html
[root@ip-172-31-1-51 R1]# ls -la /var/www/html/R3
total 4
drwxr-xr-x. 2 jenkins jenkins 24 Oct 3 05:05 .
drwxr-xr-x. 5 jenkins jenkins 36 Oct 3 03:19 ..
-rw-r--r--. 1 jenkins jenkins 56 Oct 3 05:05 index.html
[root@ip-172-31-1-51 R1]#

```

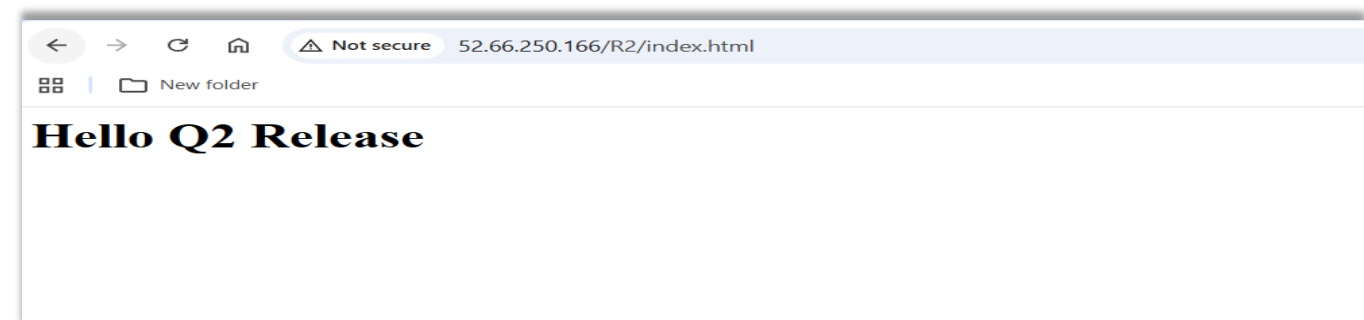
9.1 Browser Verification

Release	URL	Expected Page
Q1 Release	http://<EC2-IP>/R1/index.html	Hello Q1 Release
Q4.2 Patch	http://<EC2-IP>/R2/index.html	Hello Q4.2 Patch
Q3.3 Bug Fix	http://<EC2-IP>/R3/index.html	Hello Q3.3 Bug Fix

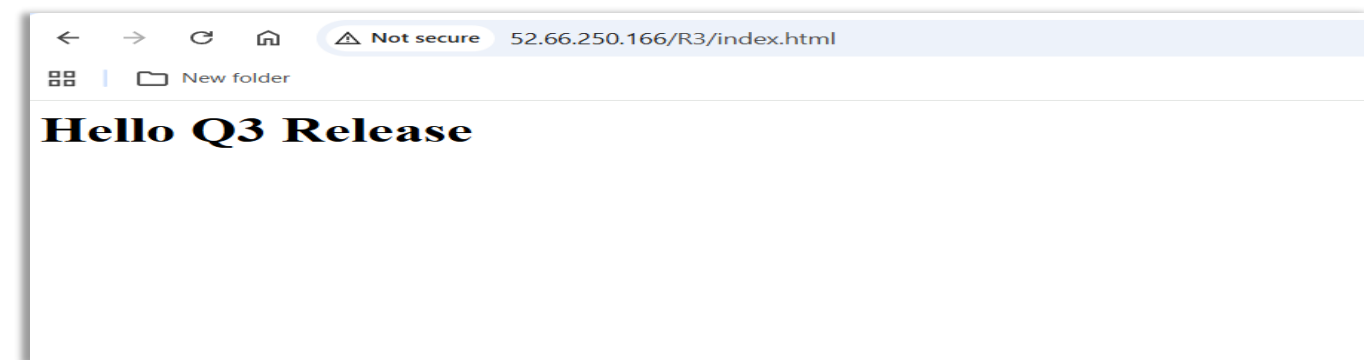
Screenshot space – R1 browser result:



Screenshot space – R2 browser result:



Screenshot space – R3 browser result:



12. Assignment Result / Report

The assignment demonstrates parallel deployment of three independent website versions on a single EC2 instance. Jenkins Freestyle Job 1 deploys the Q1 release to R1, Job 2 deploys the Q4.2 patch to R2, and Job 3 deploys the Q3.3 bug fix to R3. Apache serves all three versions simultaneously using separate URL paths.